

Project 3:

The effect of Type Ia SNe on the chemical evolution of galaxies

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ABSTRACT

The star formation histories of galaxies are expected to be affected by the feedback energy injected by Type Ia supernovae (SNIa). The progenitor for SNIa are generally separated in two cases, single-degenerated (SD) and double-degenerated (DD). According to this, SNIa are expected to occur after the formation of the stars with their different delay-times respective to each scenario. In this work we aim to test different models for supernovae progenitors using numerical simulations of isolated galaxies. Specifically we are using 4 simulations that are calibrated with different Delay-Time distributions (DTD), in order to see the effects on star formation and chemical evolution. Additionally we are using 2 simulations with different star formation efficiency (SFE), to test the effect of this property. We find importance dependences on the adopted DTD models, regulating star formation according to the SNIa rates and the timescales that each model adopt. The metallicity increase faster with higher early SNIa rates and in the α -plane we found the present of a knee for all DTD, highlighting the importance of the different timescales between SNII and SNIa. The enhanced SFE also shows a faster increasing in metallicity and a roughly constant [O/Fe] ratio.

1. Introduction.

The star formation history (SFH) of galaxies are imprinted on their stellar populations. Considering the age-metallicity degeneracy (Worthey 1994), it becomes difficult to determine the age only from properties like color or spectra. Nevertheless, element abundances ratios provide independent insights considering that different heavy elements are produced from different supernovae events, each with its corresponding timescales (Matteucci 2003).

There are two main types of supernovae; Type I and Type II supernovae (SNI and SNII). SNII correspond to the core collapsed-induced explosions of massive stars ($\geq 8M_{\odot}$) with short lifetimes ($10^6 - 10^7$ yr). SNII produce more α elements with respect to iron (Fe), relative to the solar ratios ($[\alpha/\text{Fe}] > 0$). Other core collapsed-induced explosions correspond to Type Ib and Ic Supernovae (SNIb and SNIc) but with the exception that they have lost the envelope material before the explosions. According to chemical evolution models, the contributions of SNe Ib, Ic, and HNe can be included in that of SNe II (Kobayashi et al. 2006). On the other hand, SNIa correspond to thermonuclear explosions of accreting white dwarfs (WDs) in binary systems and produce mainly Fe (Hillebrandt & Niemeyer 2000). They are also sources of kinetic-energy feedback that regulates star formation in galaxy evolution. Their remnants are the acceleration sites of cosmic rays (Freundlich & Maoz 2021), and they serve as distance indicators for cosmology (Riess et al. 1998). SNIa are widely believed to be explosions of Carbon-Oxygen (C+O) WDs whose mass has reached, by accretion, the Chandrasekhar mass limit of $1.4 M_{\odot}$ (Chandrasekhar 1931). While there is general agreement on the model of an exploding WD for SNIa, the exact nature of their binary companion is still not well understood.

Progenitor models fall into two main categories: the "single-degenerate" (SD) model, where a white dwarf increases in mass by accreting material from a main sequence binary companion (Whelan & Iben 1973), and the "double-degenerate" (DD)

model, where two WDs merge after losing energy and angular momentum to gravitational waves (Webbink 1984). Regarding this, the concept of time delay arises. The time delay refers to the delay with which Fe is ejected into the interstellar medium (ISM) by SNIa relative to the fast production of α -elements by core-collapse SNe (Pipino & Matteucci 2008). Given a starburst, SNIa events occur following a distribution of explosion times which is known as the Delay Time Distribution (DTD). The DTD is the hypothetical SN rate versus time that would follow a brief burst of star formation. The DTD is directly linked to the lifetimes, therefore, the initial masses of the progenitors and to the binary evolution timescales up to the explosion. This suggests that different progenitor scenarios predict different DTDs. Furthermore, the DTDs of different SN types dictate directly the mix of contributions of different SN types to metal enrichment throughout cosmic history (Maoz et al. 2010). We can also find empirical DTDs. Lately, Mannucci et al. (2006) proposed an empirical Bi-modal model that considers a DTD with two populations of progenitors. The first population is dominated by a fast component that explodes before 100 Myrs after the birth of the progenitor. The second late component explodes in a time range that can extend up to 10 Gyr. The shape of the distribution is given by the sum of Gaussian centered at 5×10^7 yrs, and exponentially declining function. Also, Pritchett et al. (2008) showed that the rate of SNIa explosions is about 1% of the stellar death rate, independent of star formation history, and the dependence of SNIa rate on star formation rate implied a DTD proportional to a $t^{-0.5 \pm 0.2}$ exponentially declining plateau function.

In order to understand the chemical evolution of galaxies, it becomes important to acknowledge their respective timescales. The two-infall model proposed by Chiappini et al. (1997), predicts two main episodes of gas accretion. During the first one, the bulge, halo and most of the thick disk are formed, while the second infall gave rise to the thin disk. The formation of the inner halo given by a monolithic-like collapse of gas takes about 0.8-1 Gyr and the bulge is formed in 0.1-0.5 Gyr. During this

phase part the thick disk assembles and then the innermost disk regions form due to the second infall episode. Since the inner disk formed inside-out, it takes a much longer time to form. It becomes clear that the early phases of formation are dominated by SNII (and also SNIb/c), producing most of the α -elements. The effects of SNIa start gaining importance only after 1 Gyr to begin the enrichment of the ISM (Pipino & Matteucci 2008).

Using numerical simulations we are able to test different scenarios, according to the models that are implemented for the different physical processes, such as star formation and supernovae. In simulations the metallicities are not directly calibrated with models, but they are a result from the star formation and stellar yields, mainly. Using this relevant properties of simulations we aim to study the evolution of the chemical patterns of galaxies when we consider different progenitors for SNIa and varying the star formation efficiency calibration. For this we are gonna be using isolated galaxies, to avoid considering environmental effects on the formation of the galaxies.

2. Numerical Simulations.

For the present work we are gonna be using 6 different N-body simulations of isolated galaxies (non-cosmological). They use the 3-th version of the **Galaxies with Dark matter and Gas** **intEracT** (GADGET-3) code, that consist in a TreePM smoothed particle hydrodynamics (SPH) treatment for gravity and gas hydrodynamics, respectively (Springel 2005). The code also includes a metal-dependent radiative cooling mechanism (Sutherland & Dopita 1993) and a multiphase treatment for the gas (Scannapieco et al. 2006). For star formation, chemical enrichment and feedback due to supernovae (SN) the original implementation from Scannapieco et al. (2005) have been slightly modified according to different DTDs for SNIa, as will be explain in section 2.1, and star formation efficiency (SFE). This model includes ejection of chemical elements and feedback due to SNII and SNIa, allowing the chemical evolution and the self-regulation of star formation in high temperature gas clouds due to strong metal-enhanced winds. This implementations have shown to be particularly useful in large suites of zoom-in cosmological simulations (eg. Tissera et al. 2012; Rodríguez et al. 2022).

In Figure 1 we are showing the distribution of stars (magenta) and gas (cyan) of one of the isolated galaxies, at the last snapshot. This figure highlight the capacity of the simulation to form different features, such as spiral arms in the outer region and a central spheroidal component.

The six simulated galaxies have the same initial conditions (IC) and stellar masses between $\sim 10^{10}$ and $\sim 10^{11} M_{\odot}$ (more details on the IC can be found in Jiménez et al. 2015), but DTDs and SFE have been tuned to test this models. The first four simulated galaxies have 237 snapshots and correspond to the same SFE but with different DTD models, a SD scenario (Matteucci & Recchi 2001), a DD (Greggio 2005), a Bimodal (Mannucci et al. 2006) and a power-law model (Pritchett et al. 2008). These in order to test the different origins for SNIa and their effects on chemical evolution.

For the last two simulations we consider only a DD scenario for SNIa, we want to test the effects of the star formation efficiency so we tuned the parameters of the simulation to obtain a high SFE simulation and another one with the same SFE of the previous 4 galaxies. As described in Jiménez et al. (2015), which gas particles will form stars is calibrated with the Kennicutt-Schmidt law (Kennicutt 1998; Schmidt 1959), so the tuning of the values from this relation increase the SFE by one order of

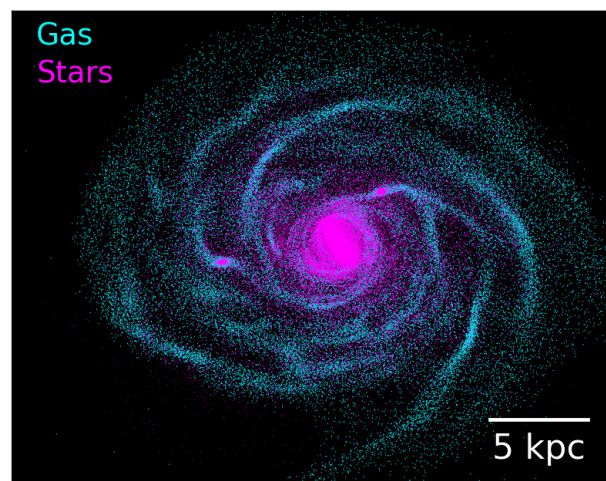


Fig. 1. Face-on spatial distribution of gas (cyan) and stars (magenta) for one of the galaxies used in this work. Relevant morphological features can be seen, such as spiral arms and a central spheroidal component.

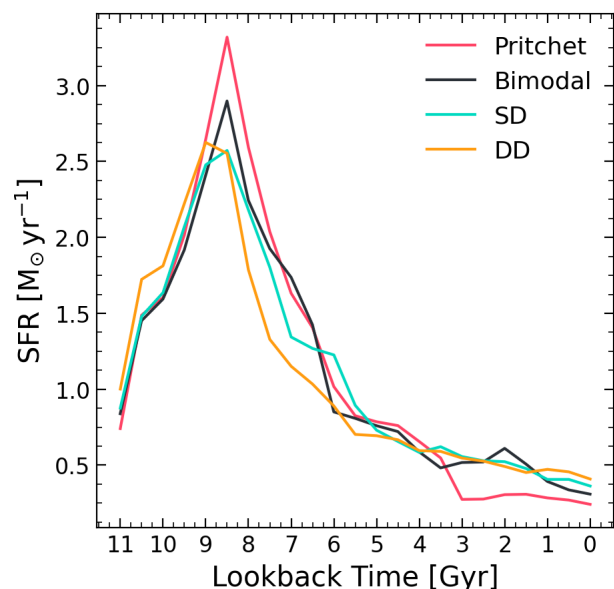


Fig. 2. SFR as a function of time for different SN progenitor models given by Pritchett (red), bimodal (black), single degenerate (cyan) and double degenerate (orange).

magnitude. The high SFE galaxy have less snapshots, 146, due to that finish star formation faster. For comparison the other galaxy will have the same amount of snapshots.

2.1. DTD implementations

The DTD in the simulations are used to efficiently track the SNIa rates in the simulation. Each particle have a formation time and according to the time that has passed since it formation there is a probability of go into a SNIa process. The DTD described the probability for the stellar particle to produce a SN and it depends on the consider progenitors. With this we can estimate the SNIa rate and limit the amount of gas particles that are receiving energy and are being enriched from SNIa. Keeping the computational efficiency and the physical background of this processes.

3. Results

3.1. Effects of Type Ia SN models

In this section, we present a comprehensive analysis of the effects of SNIa on the star formation history, age-metallicity relation, and the α -plane. Our study employs the four distinct DTD models for SNIa mentioned in Section 1.

3.1.1. Star Formation History

Figure 2 shows a high peak in the SFR for the Pritchett model at 8.5 Gyr, which aligns with the expectation of an early, intense burst of star formation. After the peak, the SFR decreases rapidly. The model suggests that the SFR peaks early in the galaxy formation and then declines as a power law. Therefore, the SFH would show a strong peak at high redshift, followed by a rapid decline. For the bimodal model, we can see that there is an initial peak at about 8.5 Gyr, that corresponds to the prompt population, and a secondary peak in at around 2 Gyr that matches the delayed supernova population, indicating ongoing star formation activities at a later period. For the SD and DD models, the plot shows a significant but less pronounced and broader peak compared to the Pritchett and Bimodal models, indicating robust star formation during that period. The SD models shows its peak at about 9 Gyr and the SFR declines more slowly than in the previous models.

The onset of the starburst in all models occurs due to the presence of available gas. Early in the history of a galaxy, the gas reservoirs might be more abundant and less metal rich compared to later epochs. The peak in SF similarity for all models can be attributed to the contribution of SNII, characterized by a smaller DTD. The decrease in SFR observed after the peak occurs when the gas is consumed and the SN feedback contribution kicks in. In order to better understand the behavior of the SFR, the visual comparison between Figure 2 and Figure A.1 provides key insights. Considering that both figures do not share the same time axis scale, three critical zones can be visually identified. The first one being the beginning of the simulation (~ 0 to 0.1 Gyr), the middle one (~ 1 Gyr), and the last one at the end of the simulation (~ 10 Gyr). We can see that models that have a bigger SN rate at earlier epochs show a slightly lower peak. Later on in the center zone, we can see that the DD model shows a more pronounced drop that relates to having the biggest SN rate at the epoch. At the end of the simulation the Pritchett model, that has the highest SN rate ends up showing again the lowest SFR value. All of these three general observations support the fact that a higher rate of SNIa at earlier times accelerates the process of feedback heating, thereby reducing the available cold gas needed for ongoing star formation, ultimately resulting in a lower SFR peak (Wang et al. 2009).

3.1.2. Age-Metallicity Relation

To study the evolution of the metallicity as a function of time we estimate the Age-Metallicity relation (AMR). To estimate the metallicity of the stars we use the $[\text{Fe}/\text{H}]$ abundances, given the fact that Fe is the main product of SNIa (Hillebrandt & Niemeyer 2000), considering the values for stellar populations with $[\text{Fe}/\text{H}] > -3.5$ and $[\text{O}/\text{Fe}] > -1$. This to avoid statistical outliers of the simulation. Another important cut was done to eliminate the stars that defined the initial metallicities of the components, masking out the stellar particles with $[\text{Fe}/\text{H}] < -2.2$ and $0.12 < [\text{O}/\text{Fe}] < 0.13$. The solar values for reference are

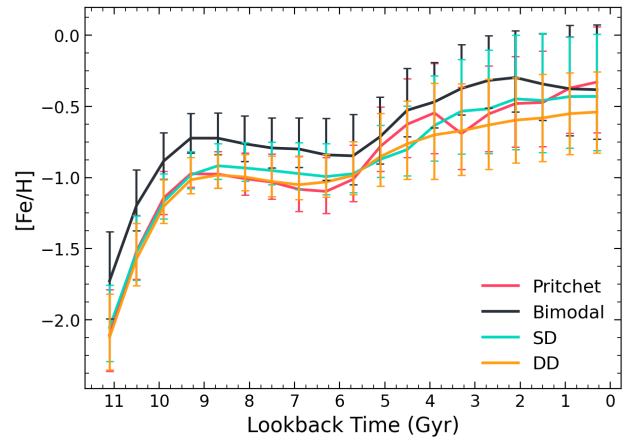


Fig. 3. Age-Metallicity for each galaxy. The distribution of colors per model corresponds to the same one in Figure 2.

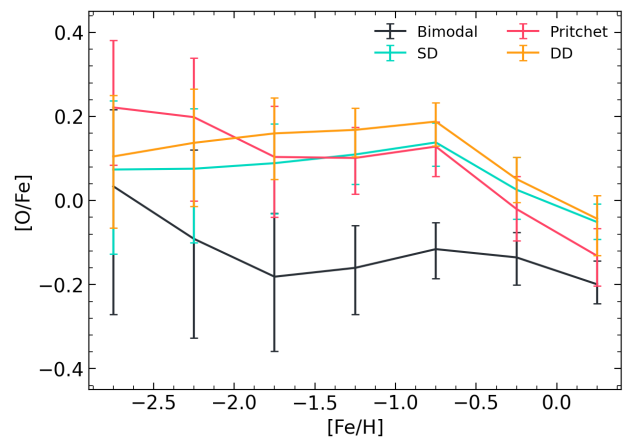


Fig. 4. Median value for the $[\text{O}/\text{Fe}]$ abundance as a function of $[\text{Fe}/\text{H}]$ for the four implemented models of SNIa. The error bars correspond to the percentile 25 and 75. The so-called knee is present for all models, but the shape and predominance of it change.

taken from Lidders (2019).

The evolution of metallicity $[\text{Fe}/\text{H}]$ across cosmic time varies consistently among different models. In early epochs (> 10 Gyr), all models exhibit low metallicity (~ -2.0), indicative of a less enriched gas from which stars formed. During intermediate epochs (9 - 10 Gyr), there is a notable increase in metallicity as the SFR arises, reflecting the enrichment of the interstellar medium by earlier stellar generations, with metallicity peaking around ~ -0.7 to -1.1 , depending on the model. Between 6-9 Gyr, a slight decrease in metallicity can be seen; this can be explained considering that galaxies often exhibit metallicity gradients, where the central regions have higher metallicity than the outskirts (Pilkington et al. 2012). Mixing processes can influence these gradients by redistributing metals from inner and outer regions, thereby forming stars based on less enriched gas, ultimately resulting in the observed decrease in metallicity.

Moving into recent epochs (< 5 Gyr), metallicity continues to arise, albeit at a slower rate, stabilizing around ~ -0.5 to -1.0 across all models with the exception of the Pritchett model that shows a small valley at about 3 Gyr. This convergence suggests a consistent level of enrichment. Over longer timescales, efficient mixing would tend to smooth out short-term fluctuations, as shown by the general trend, leading to more gradual changes

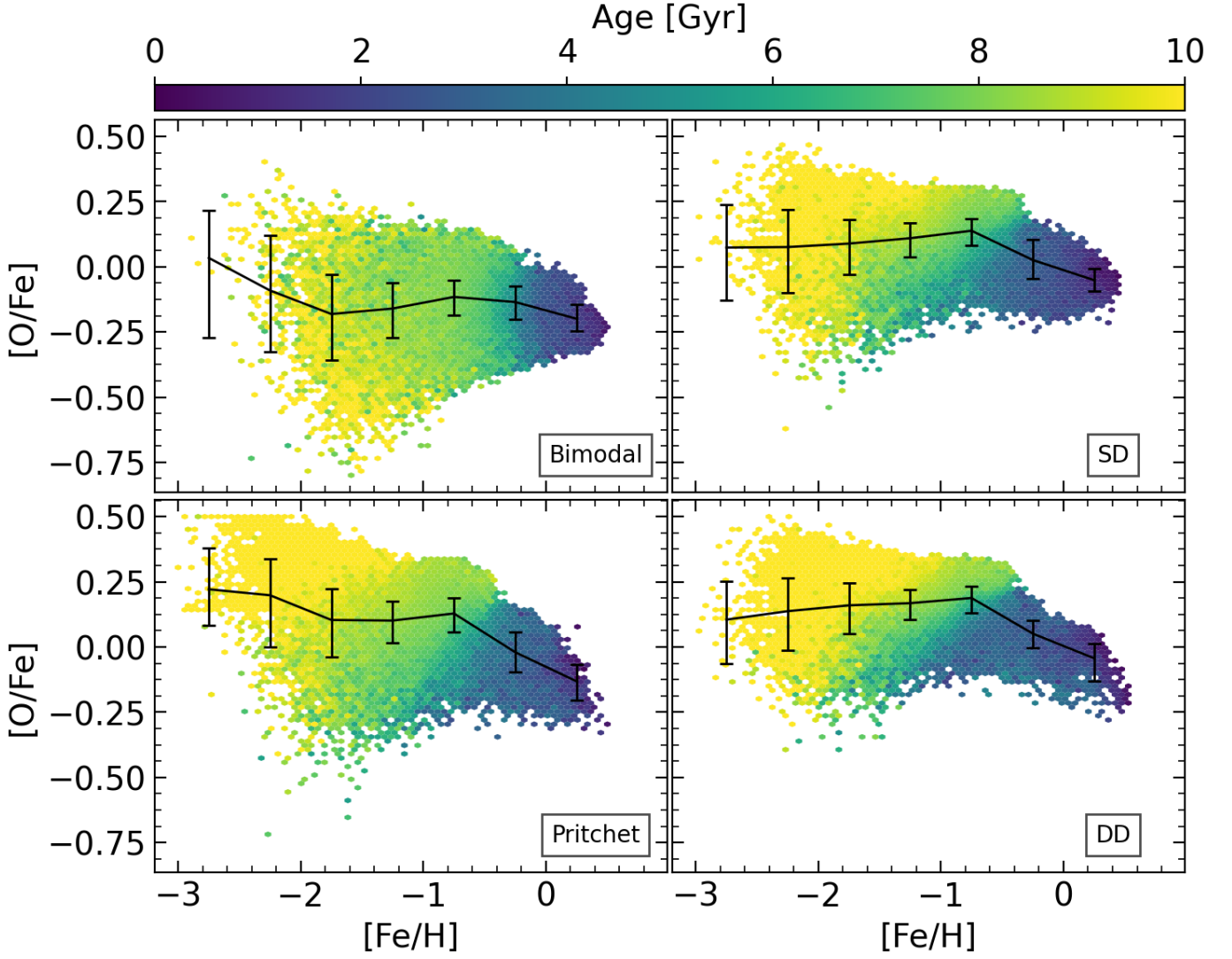


Fig. 5. Median value for the $[O/Fe]$ abundance as a function of $[Fe/H]$ for the four implemented models of SNIa. For the stellar populations an hexagonal binning have been applied, color-coded by the ages. This to quantify the effects of the models in both, chemical abundances and ages. Each model show a positive dependence with ages and $[Fe/H]$, but this dependence change with each model.

in stellar metallicity. This could be explain since the SFR at this epochs of the simulations remain relatively low, keeping the metallicity enrichment stable due to the low amount of SNIa and the smooth mixing of metals within the galaxy.

3.1.3. α -plane

To understand the different timescales of stellar enrichment that is associated to the stars in different elements, we are going to use the plane of $[O/Fe]$ vs $[Fe/H]$, also called α -plane. The $[O/Fe]$ abundance is expected to be a comparison between SNII and SNIa, since O is an α element, and is expected to be produced mainly by SNII (eg. [Bensby et al. 2013, 2014](#)). Since SNIa and SNII have different timescales ([Mannucci et al. 2006](#)), this plane can be used to establish how both abundances change with the different progenitor modelling.

In [Figure 4](#) we are comparing the four implemented models in the α -plane. All models present a decay in $[O/Fe]$ at $[Fe/H]$ higher than ~ -0.7 , the so-called knee, but how important is this decay change for the different models. This decay is expected at high metallicities because SNIa produce mainly Fe ([Hillebrandt](#)

& [Niemeyer 2000](#)), and they are still occurring after the SNII stage. This means that the oxygen abundance will not change, making the $[O/Fe]$ fraction decrease at the high-end of $[Fe/H]$. The relation of the knee and the models will be further explore with [Figure 5](#), together with the ages.

The four presented models have a different behavior in this figure probably due to the different times that takes to the SNIa to enrich the ISM, compare to the SNII. The Bimodal (black) is the model with the higher SNIa rates at earlier formation times of the stars, so as expected the Fe abundances increase very fast, making $[O/Fe]$ lower over the entire curve. On the other side, Pritchett DTD decrease exponentially with time, this is also probably producing a fast decrease in $[O/Fe]$ at lower $[Fe/H]$ values, but is less extreme compare to the Bimodal one. The SD and DD models produce a plateau at low $[Fe/H]$ values, suggesting that the predominant mechanism is the SNII at low $[Fe/H]$, and the SNIa starts to affect the plane at higher iron enrichment. The knee is also different for each model, according to the DTDs, the bimodal is expected to be the one with lower SN rates at higher delay times. This produce less Fe production of SNIa and a less steeped knee. Pritchett show the most steeped knee, this as

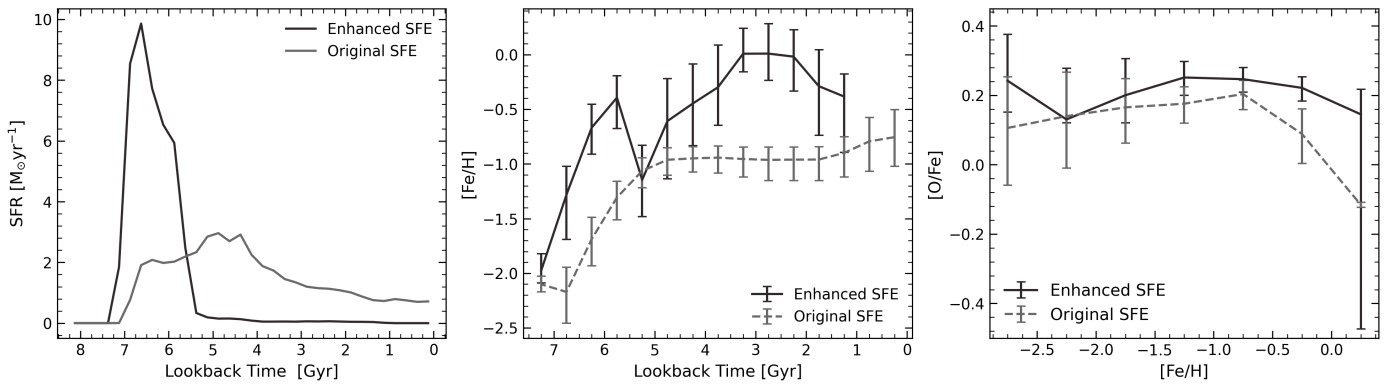


Fig. 6. Left: Star formation history of two models, an SFE enhanced (black line) and the original model (gray line), both with a DD progenitor. The enhanced model shows a very important starburst at the beginning of the simulation. Middle: Age-Metallicity relation for the two models, different $[\text{Fe}/\text{H}]$ values are reached in different timescales according to the SFE. Right: Median value for the $[\text{O}/\text{Fe}]$ abundance as a function of $[\text{Fe}/\text{H}]$ for both models. Similarly to the previous plot both models have different behaviors, that depend on the implemented model.

expected because it is the one with higher SNIa rates at younger timescales.

To better quantify the ages at which SNIa or SNIa dominates, in Figure 5 we are color coding each plane with the ages. As we can see the Bimodal model shows an increase of $[\text{Fe}/\text{H}]$ as a function of age. This enrichment occurs relatively fast with SNIa and SNIi probably occurring in closer timescales, so $[\text{O}/\text{Fe}]$ does not show a clear dependence with age. For the other three models the SNIa rates are high enough after SNIi are not occurring, to produce a decrease in $[\text{O}/\text{Fe}]$. We can notice that this decrease corresponds with the ages, with more younger stars to have, at the same $[\text{Fe}/\text{H}]$, lower $[\text{O}/\text{Fe}]$ values. The shape of this knee has been a matter of discussion in several studies, we can notice that the SD and DD progenitor models show a plateau at low and a shin at higher metallicities. This is in agreement with the 'classical knee shape from Mason et al. (2023) in different galaxies from cosmological simulations and the shape for the knee that have been also observed for the Galactic Bulge (Bensby et al. 2013).

3.2. Enhanced star formation model

Another topic that has been discussed in star formation modelling is the effects of the SFE (eg. Wang et al. 2009) on the evolution of galaxies. As we described before, in this section we are going to compare two simulated galaxies. One of them includes an enhanced star formation calibration, that will allow to compare two scenarios of formation.

The left plot of Figure 6 corresponds to the SFH for both models. The enhanced SFE one (black lines) shows an important peak of star formation at ~ 6.8 Gyr, as expected. But after this peak the star formation decreases rapidly, this suggests that the starburst depleted all the gas available to form stars, quickly. Instead, the original model reaches the peak at ~ 4.5 Gyr and has a more bursty star formation, forming stars until the end of the simulation. The middle plot of Figure 6 shows the AMR, when the enhanced SFE model is implemented we can see a much faster enrichment previous to the peak of starburst (> 5.5 Gyr). This increase is expected due to the increment of stars that can go into a SNIa process, enriching the ISM for the immediate next generation of stars (Maiolino & Mannucci 2019), compared to the other model. After the starburst, the enhanced model forms a small amount of stars that show lower metallicities at ~ 5.25 Gyr, this low metallicity suggests that these stars formed from gas in the outer regions that was not enriched in the first star-

burst. The amount of stars that forms after this period are much less (< 300 in each bin), so depending on the region that they formed the metallicities could vary a lot, as can be seen for lookback times lower than ~ 4.5 . Finally, the right plot of Figure 6 shows the α -plane of both models. The enhanced model shows a small decrease of $[\text{O}/\text{Fe}]$ at lower metallicities, probably due to the high amount of SNIa that are occurring in this part of the α -plane. This can also be seen in the AMR, where low metallicity stars are produced in the first ~ 2 Gyrs of formation. The knee is also relatively present also in the plane of the enhanced model, but the star formation is very low at the lower lookback times, so there are not enough stars to produce a relevant enrichment of SNIa (Fe), compared to the one of SNIi (O). This could explain why the knee is much less steeped.

4. Conclusions

In this study we use non-cosmological numerical simulations of isolated galaxies to test the influence of different DTD progenitor models, in order to understand their effects on galaxy formation and evolution. In specific we compare a SD, DD, Bimodal and Pritchett DTD, comparing their star formations and chemical patterns. Additionally, we study the effects of varying SFE in the DD scenario.

- i) The analysis of various star formation history (SFH) models reveals distinct patterns in the evolution of star formation rates (SFR). The Pritchett model exhibits an early, intense burst followed by a rapid decline, while the Bimodal model shows its two peaks from prompt and delayed supernova populations. The SD and DD models demonstrate significant, though less pronounced, peaks with the SD model displaying a slower decline. All models peaked at ~ 8.5 Gyr. Gas availability and metallicity play crucial roles, with early starbursts linked to abundant, less metal-rich gas. The comparison with SNIa rates plots underscores how higher rates in early epochs reduce subsequent SFR peaks due to increased feedback.
- ii) Metallicity evolution across cosmic time varies consistently among different models. In early epochs all models exhibit low metallicity (~ -2.0), indicating a less enriched gas during initial star formation. As SFR increases during intermediate epochs (9 - 10 Gyr), metallicity notably rises, peaking around -0.7 to -1.1 . A slight decrease in metallicity between 6-9 Gyr suggests the influence of metallicity gradients within

galaxies, where mixing processes redistribute metals from inner to outer regions. Moving into recent epochs (< 5 Gyr), metallicity rises stabilizing around -0.5 to -1.0 across models, except for the Pritchett model which shows a minor deviation at 3 Gyr. Over time, efficient mixing in galaxies smooths out short-term metallicity fluctuations, ensuring more gradual changes.

- iii) Regarding to the α -plane, the different models shows that there is an effect on the shape at low and high $[\text{Fe}/\text{H}]$. At low metallicities the DTDs with high SNIa rates at smaller delay times show a decrease in the $[\text{O}/\text{Fe}]$ values. This compare to the SD and DD scenarios that show the well-known plateau of the α -plane. The knee is present for all models at $[\text{Fe}/\text{H}]$ of ~ -0.7 , but how steep is the declining also depend on the model, with higher SNIa rates at larger times resulting in more steeped knees.
- iv) Increasing the SFE also show an important effect on chemical evolution of the galaxy. The short but important starburst of the enhanced model produce the increment in the amount of $[\text{Fe}/\text{H}]$ faster than the non-enhanced model. On the other way, for the $[\text{O}/\text{Fe}]$ they remained roughly constant trough all metallicities. This suggest that the enrichment by SNIa and SNIi occurred in similar timescales, when you increased the SFE.

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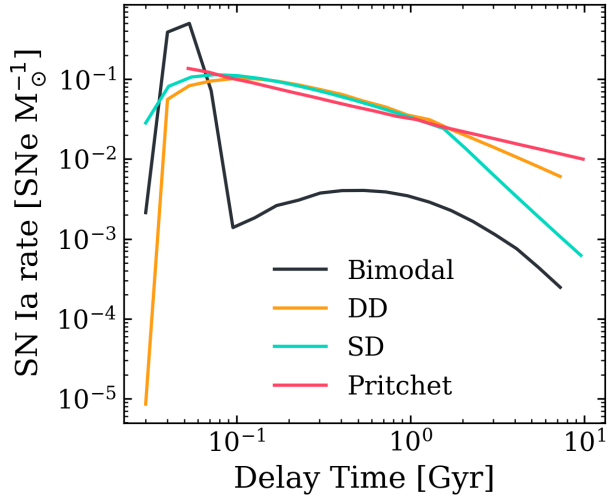


Fig. A.1. Delay Time Distribution used for each model. The different rates that dominates at different timescales can be seen here.

Appendix A: Delay Time Distributions per model